

Activity 8 Pre-Lab — Markovnikov Hydrohalogenation of Carvone

[Your name] & [partner]

[MM/DD/YY]

Organic Lab · selective HCl addition to an alkene

■ **Black** = write this in your notebook ■ **Blue** = context, don't copy it Copy the header onto every page. Draw dashed-box structures yourself.

Purpose

Add **HCl across an alkene of carvone** and show the **chemo- and regioselectivity** of the reaction: HCl adds **Markovnikov** to the isolated (isopropenyl) C=C — not the conjugated enone C=C — giving the tertiary chloride. Monitor the reaction by TLC.

DRAW — the reaction

input chemical structure of carvone → product (Markovnikov HCl adduct) **HCl is generated in situ: $\text{TMSCl} + \text{H}_2\text{O} \rightarrow \text{HCl}$. HCl adds to the isolated isopropenyl C=C (H to the CH_2 , Cl to the more-substituted carbon → 3° chloride). The conjugated ring enone C=C is unreactive.**

Chemical Reagents & Safety Table

CONTEXT ▶ Required: carvone, chlorotrimethylsilane (TMSCl), water. Typical SDS values — cross-check the SDS tile.

Chemical	Role in lab	MW	Amount used	Physical properties	Hazards (GHS)
Carvone	Substrate (the alkene, limiting)	150.22	2 mL (~1.9 g, ~0.013 mol)	Colorless–pale-yellow liquid; bp 230 °C; d 0.96 g/mL; minty smell	H315 skin irritation · H317 may cause allergic skin reaction
Chlorotrimethylsilane (TMSCl)	HCl source (reacts w/ water)	108.64	~6 mL	Colorless fuming liquid; bp 57 °C; d 0.86 g/mL; flash pt –18 °C	H225 highly flammable · H314 severe skin burns & eye damage · H332 harmful if inhaled · EUH014 reacts violently with water
Water	Hydrolyzes TMSCl → HCl (H source)	18.02	1 mL	Colorless liquid	No significant GHS hazards

FYI ▶ also used (not required): saturated $\text{NaHCO}_3(\text{aq})$ to quench/rinse the corrosive TMSCl and acid; 30% EtOAc/hexanes as the TLC solvent.

GRAB FIRST: 50-mL round-bottom + stir bar · stir plate + clamp · plastic pipettes · funnel · 10-mL graduated cylinder + 250-mL & 50-mL beakers (for TMSCI transfer) · 400-mL beaker w/ ~75 mL sat. NaHCO_3 · aqueous-waste beaker · UV-active TLC plates + capillaries + chamber (30% EtOAc/hexanes) · small vial.

Procedure Plan (left = plan, right = fill in during lab)

Plan (write before lab)

Part 1 — Set up + add HCl source

1. Weigh a **50-mL RBF**. Add **2 mL carvone** (plastic pipette), weigh again → exact mass of carvone.
2. Add **1 mL H_2O** + a stir bar; clamp over a stir plate, start stirring.
3. Have ~75 mL **sat. NaHCO_3** ready in the hood (for quenching) + a labeled aqueous-waste beaker.
4. Get **~6 mL TMSCI** from the instructor in a covered graduated-cylinder assembly. **TMSCI is corrosive & fumes — goggles + gloves; if it hits gloves/clothes, remove them immediately.**
5. Add the TMSCI to the stirring carvone/water through a funnel (keep pipette vertical). **Record the start time. TMSCI + water → HCl right in the flask, which then adds to carvone.**
6. Rinse the funnel + cylinder with sat. NaHCO_3 (it'll fizz); put rinse in aqueous waste. **Swap to fresh gloves.**

Part 2 — Monitor by TLC

7. Spot a 3-lane plate: **S** = carvone standard, **C** = co-spot (S + R together), **R** = reaction aliquot (1 drop diluted with 2–4 drops eluent). Develop in 30% EtOAc/hexanes; run first plate right away and note elapsed time.
8. Re-run TLC **every 15 min** until carvone spot is gone (~15–60 min). Record the completion time.

Part 3 — Work up **quench w/ NaHCO_3** , isolate product, IR/NMR (per procedure).

✍️ WRITE — why Markovnikov + selective

H^+ adds to give the **more stable (3°) carbocation**; only the electron-rich isolated alkene reacts, the conjugated enone is deactivated.

Experiment Notes (in lab)

Mass flask: _____ g **flask+carvone:** _____ g

Mass carvone: _____ g

TMSCI added — time: _____

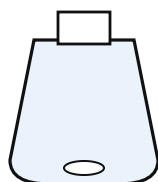
Observations on adding TMSCI (heat? fizz?):

TLC over time (R_f of S vs R each check):

0 min:
15 min:
30 min:
45+ min:

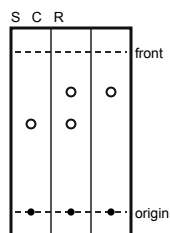
Reaction complete at: _____

Setup — draw these



50-mL RBF, stirring

reaction flask (stir)



TLC monitor (S · C · R)

CONTEXT (don't copy) ▶

Why TMSCI + water? it's a safe, controlled way to make HCl in the flask instead of pouring in hydrochloric acid.

Co-spot (C)? spotting S and R in one lane — if the reaction spot is truly new, you'll see two separated dots, confirming carvone was consumed.

Clean-up & Conclusions

- Quench acidic residues with NaHCO_3 → aqueous waste; organic product/solvents → organic waste; used capillaries → glass waste; fresh gloves after TMSCI; bench task.
- **Conclusions** (~4 sentences **after lab**): TLC evidence carvone was consumed; which alkene reacted and why (Markovnikov, more stable 3° cation, isolated vs conjugated alkene); what IR/NMR say about the product.